

Airline System

- **Idea**

- A system to manage an airline including:

- Passengers

- Flights

- Bookings

- Aircraft

- Airports

- **Tables (6)**

- Passenger

- Flight

- Booking

- Ticket

- Aircraft

- Airport

- **Relationships**

- Passenger books Flights through Booking

- Booking generates Ticket

- Flight is linked to Aircraft and Airports

- **Operations**

- Insert: add passengers, flights, bookings

- Update: modify booking or flight data

- Delete: remove records like bookings or passengers

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1 CREATE DATABASE airline_system;
2 CREATE TABLE Passenger (
3     passenger_id INT PRIMARY KEY,
4     name VARCHAR(100),
5     passport_number VARCHAR(50),
6     phone VARCHAR(20),
7     email VARCHAR(100)
8 );
9 CREATE TABLE Employee (
10     employee_id INT PRIMARY KEY,
11     name VARCHAR(100),
12     role VARCHAR(50),
13     salary DECIMAL(10,2)
14 );
15 CREATE TABLE Airport (
16     airport_id INT PRIMARY KEY,
17     name VARCHAR(100),
18     city VARCHAR(50),
19     country VARCHAR(50),
20     code VARCHAR(10)
21 );
22 INSERT INTO Passenger (passenger_id, name, passport_number, phone, email) VALUES
23 (1, 'Ahmed Ali', 'P123456', '0100000001', 'ahmed@mail.com'),
24 (2, 'Sara Mohamed', 'P234567', '0100000002', 'sara@mail.com'),
25 (3, 'Omar Hassan', 'P345678', '0100000003', 'omar@mail.com'),
26 (4, 'Mona Adel', 'P456789', '0100000004', 'mona@mail.com');
27 INSERT INTO Employee (employee_id, name, role, salary) VALUES
28 (1, 'Captain Ali', 'Pilot', 20000),
29 (2, 'Nour Ahmed', 'Crew', 8000),
30 (3, 'Hany Samy', 'Admin', 6000),
31 (4, 'Salma Hassan', 'Crew', 8500);
32 INSERT INTO Airport (airport_id, name, city, country, code) VALUES
33 (1, 'Cairo International Airport', 'Cairo', 'Egypt', 'CAI'),
34 (2, 'Dubai International Airport', 'Dubai', 'UAE', 'DXB'),
35 (3, 'JFK Airport', 'New York', 'USA', 'JFK'),
36 (4, 'Heathrow Airport', 'London', 'UK', 'LHR');
37 INSERT INTO Aircraft (aircraft_id, model, total_seats) VALUES
38 (1, 'Boeing 737-800', 180),
39 (2, 'Airbus A320', 170),
40 (3, 'Boeing 777-300', 300),
41 (4, 'Airbus A380', 500);

```

```

22 CREATE TABLE Aircraft (
23     aircraft_id INT PRIMARY KEY,
24     model VARCHAR(50),
25     total_seats INT
26 );
27 CREATE TABLE Flight (
28     flight_id INT PRIMARY KEY,
29     flight_number VARCHAR(20),
30     departure_airport_id INT,
31     arrival_airport_id INT,
32     departure_time DATETIME,
33     arrival_time DATETIME,
34     aircraft_id INT,
35     FOREIGN KEY (departure_airport_id) REFERENCES Airport(airport_id),
36     FOREIGN KEY (arrival_airport_id) REFERENCES Airport(airport_id),
37     FOREIGN KEY (aircraft_id) REFERENCES Aircraft(aircraft_id)
38 );
39
40 CREATE TABLE Ticket (
41     ticket_id INT PRIMARY KEY,
42     passenger_id INT,
43     flight_id INT,
44
45     INSERT INTO Flight (flight_id, flight_number, departure_airport_id, arrival_airport_id, departure_time, ar
46 (1, 'MS101', 1, 2, '2026-05-01 10:00:00', '2026-05-01 14:00:00', 1),
47 (2, 'MS102', 2, 1, '2026-05-02 12:00:00', '2026-05-02 16:00:00', 2),
48 (3, 'MS103', 1, 3, '2026-05-03 08:00:00', '2026-05-03 18:00:00', 3),
49 (4, 'MS104', 3, 1, '2026-05-04 09:00:00', '2026-05-04 19:00:00', 1);
50 INSERT INTO Ticket (ticket_id, passenger_id, flight_id, seat_number, price, booking_date) VALUES
51 (1, 1, 1, 'A1', 2000, '2026-04-20'),
52 (2, 2, 2, 'B2', 2500, '2026-04-21'),
53 (3, 3, 3, 'C3', 3000, '2026-04-22'),
54 (4, 1, 2, 'A2', 2200, '2026-04-23');
55
56 UPDATE Ticket SET seat_number = 'A10' WHERE ticket_id = 1;
57 UPDATE Ticket SET price = 2600 WHERE ticket_id = 2;
58 UPDATE Ticket SET booking_date = '2026-04-25' WHERE ticket_id = 3;
59 UPDATE Ticket SET passenger_id = 2 WHERE ticket_id = 4;
60
61 DELETE FROM Ticket WHERE ticket_id = 4;

```

```

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    salary DECIMAL(10,2)
);
CREATE TABLE Airport (
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    aircraft_id INT PRIMARY KEY,
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    flight_id INT PRIMARY KEY,
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    departure_airport_id INT,
    arrival_airport_id INT,
    departure_time DATETIME,
    arrival_time DATETIME,
    aircraft_id INT,
    FOREIGN KEY (departure_airport_id) REFERENCES Airport(airport_id),
    FOREIGN KEY (arrival_airport_id) REFERENCES Airport(airport_id),
    FOREIGN KEY (aircraft_id) REFERENCES Aircraft(aircraft_id)
);
CREATE TABLE Ticket (
    ticket_id INT PRIMARY KEY,
    passenger_id INT,
    flight_id INT,
    seat_number VARCHAR(10),
    price DECIMAL(10,2),
    booking_date DATETIME,
    FOREIGN KEY (passenger_id) REFERENCES Passenger(passenger_id),
    FOREIGN KEY (flight_id) REFERENCES Flight(flight_id)
);

```

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- (2, 'Sara Mohamed', 'P2 34567', '0100000002', 'sara@mail.com'),
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- INSERT INTO Flight (flight_id, flight_number, departure_airport_id, arrival_airport_id, departure_time, arrival_time, aircraft_id) VALUES
- (1, 'MS101', 1, 2, '2026-05-01 10:00:00', '2026-05-01 14:00:00', 1),
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- DELETE FROM Ticket WHERE ticket_id = 4;